CISC 7406 PROJECT PROPOSAL
TO THINK OR NOT TO THINK: MAPPING THE CHAIN-OF-THOUGHT BENEFIT BOUNDARY ACROSS SERIAL DEPTH AND OUTPUT CONSTRAINTS

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ABSTRACT

Chain-of-Thought (CoT) helps on mathematical reasoning but can degrade instruction following. We hypothesize one surface behind both: CoT's net gain should rise with an item's serial computation depth (d) and fall with its output constraint count (k). On a controlled synthetic grid we will locate the boundary $d^*(k)$ where that gain vanishes, and test it against work-matched serial/parallel controls, GSM8K and IFEval anchors, and a decomposition of the joint gain into computation versus format. We then ask whether the model actually reads the trace, by shuffling solution steps, and whether the required length follows $L^* = \tau d + ck$, with length a mediator rather than a cause. Together these give a per-item rule for deciding, before generation, whether to think at all.

1 RESEARCH QUESTION

Can we decide, before any generation, whether asking a model to write an explicit reasoning trace will improve its answer? Prior work finds CoT gains concentrated in mathematical and symbolic tasks (Sprague et al., 2024), yet explicit reasoning can impair instruction following (Li et al., 2025). We attribute both to two task-level properties: **serial depth** d, the critical-path length of the computation dependency graph, and **constraint count** k, the number of output requirements. We will test whether larger d raises CoT's relative benefit, whether larger k offsets it, and whether the two interact to give a predictable boundary $\Delta(d, k) = 0$.

We will sweep the full $d \times k$ space on a controlled synthetic dataset, with GSM8K anchoring the depth axis and IFEval the constraint axis at near-zero depth. Three further tests will guard the boundary against alternative explanations: work-matched serial/parallel controls (separating serial structure from total work), an exact decomposition of the joint gain into computation, format, and simultaneous parts, and step shuffling (dependency-based rather than position-based reading). A budget sweep will ask whether the minimal sufficient reasoning length follows $L^*(q) = \tau d + ck$, i.e. whether constraints lengthen traces without adding effective computation. Together these will show when CoT helps, when it hurts, and whether one predictive account covers both.

2 RELATED WORK

CoT improves multi-step reasoning by generating intermediate computations Wei et al. (2022). Sprague et al. Sprague et al. (2024) show, across 20 datasets and 14 models, that its benefits concentrate in mathematical and symbolic tasks. Theoretical studies further connect CoT to inherently serial computation and increased Transformer expressiveness Li et al. (2024).

However, explicit reasoning also introduces costs. Li et al. Li et al. (2025) report reduced instruction-following accuracy in 13 of 14 models on IFEval, associated with diminished attention to output constraints. Moreover, reasoning accuracy may follow an inverted-U relationship with reasoning length Wu et al. (2025), while excessive test-time computation yields diminishing or negative returns Zhou et al. (2026).

Together, these findings suggest that CoT's usefulness depends on both computational demands and output requirements. We therefore investigate two interpretable task properties: **serial depth** d and **output constraint count** k. Using controlled synthetic tasks and complementary validation on GSM8K and IFEval, we aim to identify a potential decision boundary $\Delta(d, k) = 0$ separating beneficial from harmful reasoning.

3 PLANNED APPROACH

3.1 TASK QUANTITIES

Serial depth d is the critical-path length of the dependency graph over a problem's reference operations, **width** w the number of independent intermediate values that must all be produced and later consumed, and **constraint count** k the number of machine-checkable output requirements. Depth requires an order forced by data dependence, not by bracketing, so every chain we use changes value when its steps are reordered.

3.2 DESIGN

We sweep the full $d \times k$ grid on synthetic tasks, running direct answering and CoT on every item with Qwen3-4B and scoring answer correctness and constraint satisfaction separately. A logistic model with $\mathrm{cot} \times d$ and $\mathrm{cot} \times k$ interactions locates the boundary $\Delta(d, k) = 0$ in the joint success rate.

Because total work could also explain a depth effect, we match serial and parallel variants on W: $W = d$ for a serial chain and $W = w + (w - 1)$ for a width-w variant. We compare error-reduction rates, and read the contrast as joint rather than as separate estimates of d.

GSM8K anchors the depth axis ($k = 0$) and IFEval the constraint axis ($d \approx 0$). Step shuffling tests whether the trace is read along the dependency chain, and a budget sweep gives the minimal sufficient length $L^* = \tau d + ck$.

3.3 GSM8K AND SYNTHESIS DATASET

GSM8K provides 8,500 grade-school word problems whose gold solutions annotate every calculation as `<<48/2=24>>`; 98.7% of items therefore yield a dependency graph automatically, and we take serial depth to be its critical path ($d \in [2, 6]$, median 3). Because real problems span such a narrow depth range and carry no constraints, we generate synthetic items with d, w and k set by construction, exact ground truth, and work matched across serial and parallel variants.

3.4 PRELIMINARY STUDY

Before the full study we ran a small pilot on Qwen3-1.7B. Two facts emerged. GSM8K gold solutions annotate every calculation, so 98.7% of items yield an automatically parsed dependency graph whose critical path gives serial depth. And on a synthetic dataset with depth and parallelism set by construction, the chain-of-thought gain concentrated on serial items ($\Delta \approx +0.42$ to $+0.65$) and vanished on parallel items of comparable work, supporting our premise that the relevant variable is serial depth rather than task difficulty.

4 EXPECTED FINDING

We expect CoT to become more beneficial as serial computation depth d increases, but less beneficial as the number of output constraints k grows. We predict that these competing effects will create **decision boundary** $\Delta_J(d, k) = 0$, separating tasks where CoT improves joint success from those where direct answering is preferable.

Controlled synthetic tasks will allow us to estimate this boundary, while GSM8K and IFEval will provide complementary validation of the depth-related and constraint-related trends. We further expect work-matched serial tasks to benefit more from CoT than parallel tasks, suggesting that computational structure matters beyond total workload.

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